
Text Mining & Search: Topic Modelling and Text Summarisation

Elena Maggiore, Marco Rosato

Milano Bicocca, Data Science

SUMMARY

This project aims to perform topic modelling and text summarisation over the *Wikipedia Featured Articles* corpus to generate summaries that resemble human-written references. We implemented and compared extractive and abstractive methodologies, evaluating them via ROUGE, BERTScore, and SummaC. For topic modelling, we compared Probabilistic Latent Semantic Analysis (pLSA), Latent Dirichlet Allocation (LDA), and BERTopic. Our results indicate that pLSA offered the best topic interpretability for this dataset. In summarisation, abstractive models (specifically Large Language Models) achieved the highest linguistic quality, while extractive methods proved superior for factual consistency.

1. INTRODUCTION & DATASET

Text summarisation is a crucial tool for condensing long documents while retaining essential meaning. This project integrates topic modelling into an **extractive summarisation framework** to score sentence importance and compares this against **abstractive summarisation** techniques.

1.1 Dataset Description

The corpus selected for this study is the “*Wikipedia Featured Articles*” dataset, originally curated by the Multiling Community for the Headline Generation Task ^[8]. Although the complete dataset contains over 13,000 articles across 42 languages, this project focuses exclusively on the English subset, resulting in a training set of 3,793 documents and a test set of 30 documents.

We selected this specific corpus for three strategic reasons. First, “Featured Articles” represent the **highest standard of Wikipedia content**, ensuring that our input data possesses optimal linguistic quality and structure. Second, because the articles are randomly selected from the pool of featured content, **they cover a diverse range of technical fields**, preventing our topic models from over-fitting to a specific domain (e.g., only sports or only history). Finally, **each document includes a human-authored summary**. These summaries, originally written by the article creators, serve as a Gold Standard against which we can measure the performance of our extractive and abstractive models.

1.2 Data Loading & Cleaning

The raw dataset was stored in XML format. We utilized the Python *ElementTree* library to parse the document structure, extracting the Title, Summary, and Body for each article. The Body was further processed to flatten the nested section structure into a continuous text stream for analysis.

Exploratory Data Analysis

Before applying any modelling techniques, we performed an exploratory analysis to understand the structural characteristics of the corpus. Specifically, we analyzed the distribution of document lengths (word count) and the complexity of the articles (number of sections).

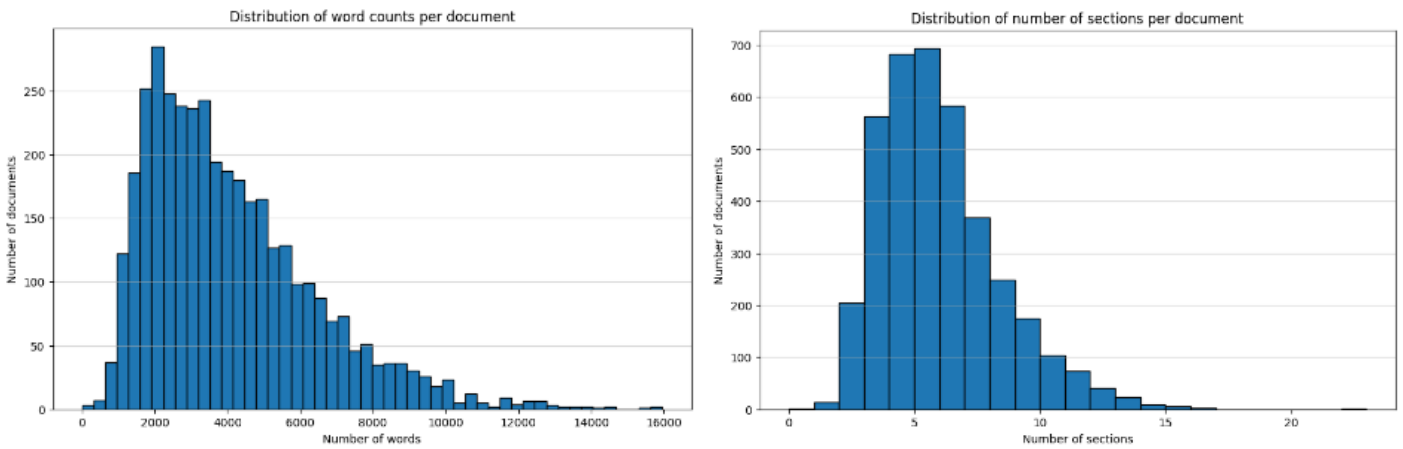


Figure 1: Distribution of word counts (left) and number of sections (right) per document.

As illustrated in **Figure 1**, both distributions are right-skewed. While the majority of documents contain between 1000 and 5000 words, a tail extends up to 16000 words. Similarly, while most articles have around 5 sections, outliers possess over 20.

Cleaning Pipeline

Since the text is from Wikipedia, it may contain noise that requires filtering. We applied a general cleaning function using *BeautifulSoup* and regular expressions to remove:

- HTML tags and encoded symbols (e.g., &).
- Wikipedia-specific reference markers (e.g., [1], [citation needed]).
- Non-ASCII characters and irregular whitespace patterns.

Topic Modelling Pre-processing

For the probabilistic models (pLSA and LDA), we implemented an additional pipeline to ensure the semantic quality of the resulting topics. Beyond standard **tokenisation, lowercasing, removal of punctuation and stop words**, we focused on **Entity Preservation via Bigram and Trigram** generation using *gensim*. We also applied *WordNetLemmatizer* to reduce the vocabulary size by mapping words to their **dictionary root**, ensuring that morphological variations were treated as identical features during the matrix factorization process.

2. TOPIC MODELLING

To support our extractive summarisation framework, we first needed to identify the latent thematic structures within the corpus. By modelling these topics, we can later score sentences based on how well they represent the core themes of a document. We implemented and compared three distinct techniques: **Probabilistic Latent Semantic Analysis (pLSA)** ^[1], **Latent Dirichlet Allocation (LDA)** ^[2], and **BERTopic** ^[3].

2.1 Methodology

We employed two statistical approaches and one deep-learning approach to determine which yielded the most interpretable topics for this specific dataset.

- **Probabilistic Latent Semantic Analysis (pLSA):** We implemented pLSA utilizing Non-Negative Matrix Factorisation (NMF) with the Kullback-Leibler divergence objective. This method approximates the decomposition of the Document-Term matrix into **Document-Topic** and **Topic-Word** matrices. We utilized *TfidfVectorizer* to generate the input matrix.
- **Latent Dirichlet Allocation (LDA):** LDA introduces Dirichlet priors (α , β) to control the sparsity of topic distributions. We utilized the standard *scikit-learn* implementation, treating documents as a **mixture of topics** and topics as a **mixture of words**.
- **BERTopic:** Unlike the previous two, BERTopic leverages semantic embeddings. We utilized *jina-embeddings-v2-small-en* (chosen for its 8192-token limit) to convert documents into **dense vectors**. We then applied **UMAP** for dimensionality reduction and **HDBSCAN** for clustering, followed by a class-based TF-IDF (**c-TF-IDF**) to extract topic representations. This allowed us to test if semantic clustering could outperform statistical word counting.

We implemented and compared three different topic modelling techniques: **Probabilistic Latent Semantic Analysis (pLSA)**, **Latent Dirichlet Allocation (LDA)**, and **BERTopic**. While pLSA and LDA rely on statistical word counts, BERTopic leverages semantic embeddings. We evaluated these models to select the technique and number of topics that yielded the highest coherence.

2.2 Model Tuning & Evaluation

We did not rely on default parameters; instead, we conducted a grid search to optimize the hyperparameters for both pLSA and LDA. We varied the *min_df* and *max_df* thresholds to control vocabulary size, filtering out noise and remaining stop words. We evaluated the models based on Topic Coherence (C_v), which measures the semantic similarity between the top words in a topic.

Statistical Model Results

For pLSA, we observed a distinct trade-off between **Reconstruction Error** and **Coherence**. As the number of topics (k) increased, the reconstruction error decreased (indicating a better fit to the training data), but coherence often plateaued, suggesting the onset of overfitting.

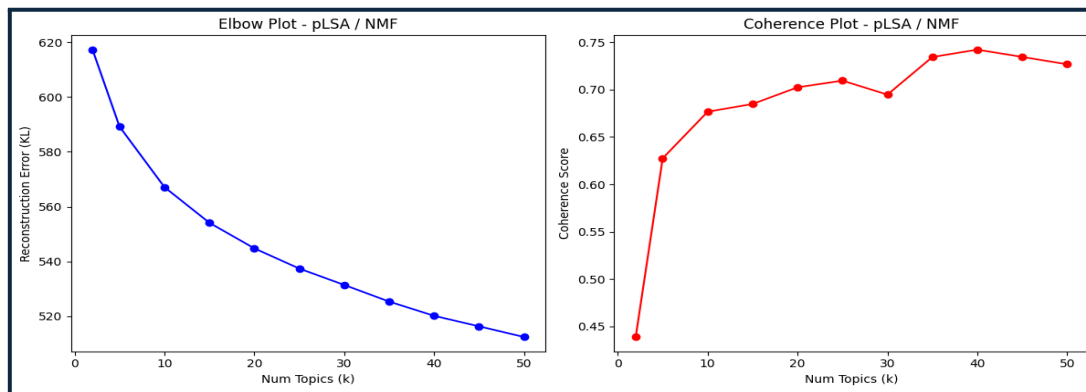


Figure 2a: Evaluation metrics for pLSA. We selected $k=40$ based on the Coherence peak.

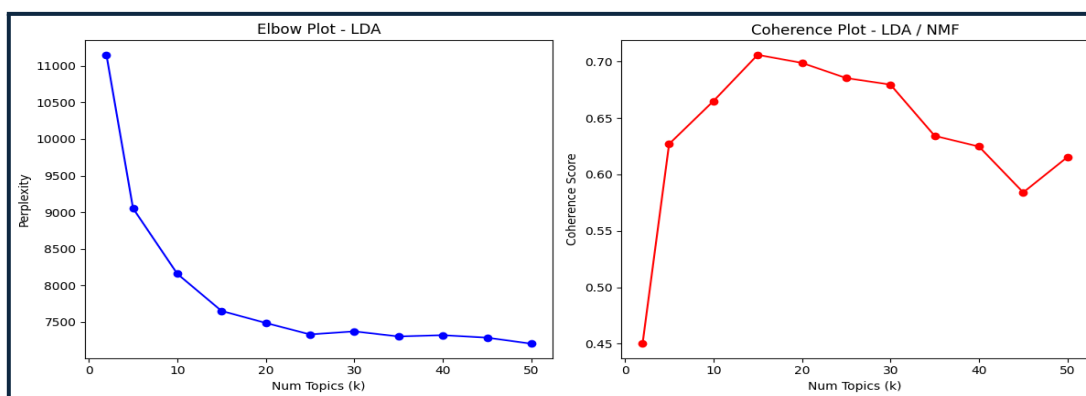


Figure 2b: Evaluation metrics for LDA. We selected $k=15$ based on the Coherence peak.

As shown in **Figure 2a and 2b**, pLSA achieved its peak coherence of **0.7421** at **40** topics. LDA performed slightly worse, peaking at **0.7059** with **15** topics.

BERTopic Results

While BERTopic excels at capturing context, our analysis of the Similarity Matrix revealed consistently high similarity scores across the topic set. This is due to the prevalence of generic, polysemous terms that appeared across multiple clusters. Furthermore, standard coherence metrics (C_v) rely on strict token co-occurrence probabilities, penalizing embedding-based models like BERTopic. Consequently, BERTopic yielded a significantly lower coherence score (**0.6237**) compared to the statistical methods.

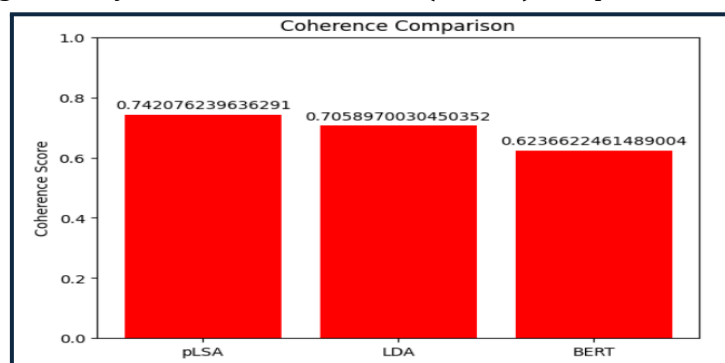


Figure 3: Comparison of best Coherence Scores across all three models.

2.3 Selected Model Analysis

Based on the results in **Figure 3**, **pLSA (40 topics)** was selected as the optimal model. It offered the best balance of topic compactness and interpretability.

To validate the quality of this model, we first examined the Topic Prevalence across the training corpus (Figure 4). Ideally, topics should be distributed somewhat evenly.

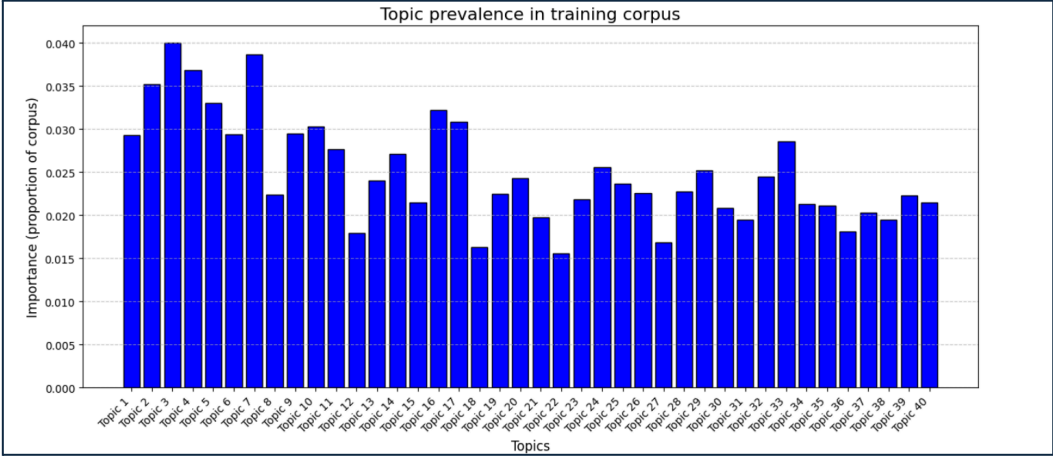


Figure 4: Topic prevalence for the best pLSA model. The distribution is relatively balanced.

As observed in **Figure 4**, the prevalence is stable across the 40 topics. No topic falls below a 1.5% share of the corpus, and the most relevant themes (Topics 2, 3, 4, and 7) appear in over 3.5% of documents. This confirms that the model successfully disentangled distinct themes rather than collapsing them into a single generic cluster.

Finally, to visualize the semantic relationships between these documents, we generated a t-SNE projection of the document-topic distribution (**Figure 5**).

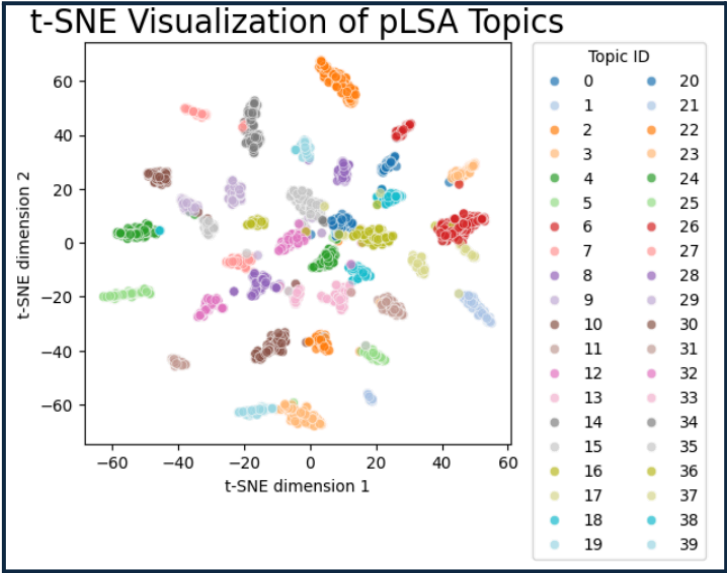


Figure 5: t-SNE visualization of the 40 topics. Distinct clusters indicate clear semantic separation.

The t-SNE projection shows distinct clusters (representing clear topics) alongside natural overlap. This structure accurately reflects the encyclopedic nature of the dataset, where topics often intersect (e.g., a "History" article may also contain "Political" terms). We concluded that pLSA provided the most robust foundation for the subsequent extractive summarisation task.

3.TEXT SUMMARISATION

The core objective of this project was to generate single-document summaries for the 30 test articles. We compared two fundamental approaches: **Extractive**, which selects salient sentences directly from the source, and **Abstractive**, which generates new text to capture the "gist" of the document.

3.1 Extractive Summarisation

Our extractive system relies on **Topic Representation**. Instead of simple word frequency, we score sentences based on how well they represent the latent topics identified in the previous phase.

Methodology

The process follows a three-step pipeline:

1. **Matrix Decomposition:** The document is represented as a TF-IDF matrix (X). This matrix is decomposed into a **Sentence-Topic matrix** (W) and a **Topic-Word matrix** (H).
2. **Sentence Scoring:** We treat sentences as vectors in the latent topic space. The "importance" of a sentence is calculated as its **vector norm** (its distance from the origin). Sentences with a high norm explain the topics best
3. **Selection via MMR:** To avoid redundancy (e.g., selecting two nearly identical sentences), we implemented **Maximal Marginal Relevance (MMR)**. This algorithm iteratively selects sentences that have high importance scores but low similarity to sentences already included in the summary.

Global vs Local Approaches

We implemented two variations of this algorithm to test the impact of prior knowledge:

- **Global Extractive (Corpus-Aware):** We utilized the **pLSA model (40 topics)** trained on the entire training corpus to initialize the matrices. This approach forces the summariser to recognize themes that are globally important across Wikipedia.
- **Local Extractive (Document-Specific):** The algorithm was fed *only* the single test document. Topics were inferred "dynamically" for that specific text by minimizing Kullback-Leibler divergence locally. This allows the model to capture very specific sub-themes unique to that document, without influence from the external corpus.

3.2 Abstractive Summarisation

Abstractive models offer higher fluency and better narrative coherence but introduce the risk of "hallucination" (generating unsupported content). We tested two architectures designed for long-context documents.

Longformer Encoder-Decoder (LED)

Standard Transformers (e.g., BERT) are limited to 512 tokens, which is insufficient for our featured articles. We utilized the **Longformer Encoder-Decoder (LED)** ^[7], a transformer optimized for long sequences (up to 16,384 tokens).

- **Mechanism:** LED uses a sparse attention mechanism (sliding window) to process local context while retaining global attention on specific tokens.
- **Implementation:** We utilized a version of the LED model that was already fine-tuned for summarisation tasks (specifically on news corpora), applying it directly to our data without further training. To prevent the model from producing truncated outputs, we applied a post-processing heuristic that trims the generated text to the last complete sentence delimiter.

Large Language Model (LLM)

We utilized **Qwen2.5-Instruct (7B)** ^[9], a decoder-only model fine-tuned for instruction following. Running a 7-billion parameter model on a Google Colab T4 GPU presents memory constraints. To summarize documents exceeding the GPU's context window, we implemented a **Chunking and Map-Reduce strategy**:

1. **Chunking:** The input document is segmented into blocks of 512 tokens. To preserve semantic integrity, we implemented a mechanism that ensures splits occur only at sentence boundaries.
2. **Map (Local Summarisation):** Each chunk is passed to the LLM with a specific instruction prompt to extract key information. We utilized **4-bit quantization (NF4)** to fit the model weights into memory.
3. **Reduce (Global Summarisation):** The resulting intermediate summaries are concatenated. If this concatenation still exceeds the context limit, the process repeats.
4. **Final Pass:** The aggregated text is fed into the LLM one last time to produce a cohesive, narrative limited to 300 tokens. We **disabled sampling** to minimize randomness and potential hallucinations.

4. EVALUATION

Evaluating generated summaries is challenging, as there is no single "correct" summary since the ground truth is human-written. To ensure a comprehensive assessment, we utilized three distinct metrics that evaluate different aspects of quality: lexical overlap, semantic similarity and factual consistency.

4.1 Evaluation Metrics

1. **ROUGE (Recall-Oriented Understudy for Gisting Evaluation)** ^[4]: We evaluated three variants:
 - **ROUGE-1:** Unigram (single word) overlap.
 - **ROUGE-2:** Bigram overlap.
 - **ROUGE-L:** Longest Common Subsequence.
2. **BERTScore:** To address ROUGE's limitations, we used BERTScore ^[5]. This metric computes similarity using contextual embeddings, rewarding models that capture the correct meaning even if the wording differs from the reference.
3. **SummaC (Summary Consistency):** Unlike the previous two, SummaC ^[6] does not use the human reference. Instead, it employs a Natural Language Inference (NLI) model to verify if the summary is factually entailed by the source document. This is critical for detecting hallucinations.

4.2 Comparative Results

Table 1 presents the consolidated performance of all four methodologies.

Technique	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore(F1)	SummaC
Global Extractive	0.23	0.05	0.20	0.8321	0.9437
Local Extractive	0.24	0.06	0.20	0.8316	0.9344
Abstractive (LED)	0.20	0.05	0.18	0.8175	0.8968
Abstractive (LLM)	0.26	0.07	0.23	0.8498	0.5089

Table 1: Performance Comparison across all metrics

4.3 Discussion & Analysis

Fluency and Semantic Similarity

The **LLM (Qwen2.5)** emerged as the clear winner in terms of writing quality, achieving the highest scores across all ROUGE and BERTScore metrics. While the generally low ROUGE-2 values (0.05–0.07) across the board highlight the difficulty of matching human phrasing exactly, the LLM still performed better than the extractive methods. This suggests that the generative model is better equipped to replicate the natural, abstractive style of the “Gold Standard” references, whereas extractive models are limited by the rigid vocabulary of the source text. The high BERTScore (**0.8498**) further confirms that even when the LLM uses different wording, it captures the semantic essence of the reference summary more effectively than the statistical approaches.

The Hallucination Trade-off

A critical insight is the discrepancy in factual grounding (SummaC).

- **Global Extractive** achieved the highest consistency (**0.9437**). By mechanically selecting sentences from the source, it guarantees that 94% of the summary is directly supported by the text.
- **The LLM** scored remarkably low (**0.5089**). This stark drop is likely due to "extrinsic hallucinations": facts that are true in the real world (and known to the LLM via pre-training) but were not present in the specific source text. Since SummaC measures entailment strictly against the provided input, it correctly identifies these external additions as "unsupported."

Conclusion on Model Selection

The results present a dichotomy. If the goal is to produce a summary that reads naturally, the **Abstractive LLM** is superior. However, for an automated system summarizing encyclopedic documents where information reliability is paramount, the **Global Extractive Summarization via pLSA** is the most robust solution. It offers the best balance of high semantic similarity (BERTScore 0.83) and guaranteed factual consistency.

5. CONCLUSION

This study explored the interaction between Topic Modelling and Text Summarisation on high-quality encyclopedic text.

Topic Modelling Selection

Our analysis suggests that for a corpus of long, diverse documents, statistical approaches still hold significant value. Probabilistic LSA (pLSA) proved to be the most effective model, achieving the highest Coherence Score (0.7421) with 40 topics. While semantic approaches like BERTopic capture subtle nuances, they struggled to form distinct, interpretable clusters in this specific high-dimensional space, often suffering from semantic drift.

Summarisation Performance & The "Readability vs. Reliability" Trade-off

Our results highlight a fundamental trade-off in automated summarisation:

1. **The LLM (Abstractive)** is the superior choice for **Linguistic Quality**. It naturally mimics human writing styles, achieving the highest ROUGE and BERTScore values.
2. **Global Extractive (pLSA-based)** is the superior choice for **Factual Consistency**. It achieved a high SummaC score (0.9437), making it the safest option for information-critical tasks.

Limitations of the Extractive Approach

However, the reliability of the Extractive model comes at the cost of narrative cohesion. While the sentences are factually accurate, the resulting summary often lacks a natural flow. Because the algorithm mechanically selects sentences based on statistical scores, it frequently omits necessary context, such as introductory clauses or the antecedents of pronouns. This results in a "disjointed" reading experience where the summary feels like a list of isolated facts rather than a story, forcing the reader to infer the connections between sentences.

FUTURE WORK

To address the trade-off between factual reliability and narrative coherence, future research could investigate a **two-stage "Extract-then-Abstract" framework**. This hybrid architecture would utilize the Global Extractive model as a content selector to identify factually supported sentences (preserving the high SummaC score). This would be followed by an LLM constrained to act solely as a stylistic paraphraser to merge these sentences into a cohesive narrative. By grounding the LLM's generation in pre-verified extractive content, this approach aims to combine the factual precision of statistical modelling with the linguistic fluency of generative AI.

6. REFERENCES

1. **Hofmann, T. (1999).** Probabilistic Latent Semantic Indexing. Proceedings of the 22nd Annual International ACM SIGIR Conference on Research and Development in Information Retrieval.
2. **Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003).** Latent Dirichlet Allocation. Journal of Machine Learning Research, 3, 993-1022.
3. **Grootendorst, M. (2022).** BERTopic: Neural topic modeling with a class-based TF-IDF procedure. arXiv preprint arXiv:2203.05794.
4. **Lin, C.-Y. (2004).** ROUGE: A Package for Automatic Evaluation of Summaries. Text Summarization Branches Out.
5. **Zhang, T., Kishore, V., Wu, F., Weinberger, K. Q., & Artzi, Y. (2020).** BERTScore: Evaluating Text Generation with BERT. ICLR 2020.
6. **Laban, P., Schnabel, T., Bennett, P. N., & Hearst, M. A. (2022).** SummaC: Re-Visiting NLI-based Models for Inconsistency Detection in Summarization. Transactions of the Association for Computational Linguistics.
7. **Beltagy, I., Peters, M. E., & Cohan, A. (2020).** Longformer: The Long-Document Transformer. arXiv preprint arXiv:2004.05150.
8. **Litvak, M., Conroy, J., & Rankel, P. (2019).** The MultiLing 2019 Headline Generation Task. Proceedings of the Workshop on MultiLing 2019.
9. **Yang, A., et al. (2024).** Qwen2.5 Technical Report. *arXiv:2412.15115*